Social Science & Medicine 191 (2017) 237–250

Contents lists available at ScienceDirect

Social Science & Medicine

journal homepage: www.elsevier.com/locate/socscimed

Review article

Racial residential segregation and adverse birth outcomes: A systematic review and meta-analysis

Renee Mehra*, Lisa M. Boyd, Jeannette R. Ickovics

Yale University, USA

ARTICLE INFO

Article history:
Received 2 February 2016
Received in revised form
25 August 2017
Accepted 8 September 2017
Available online 11 September 2017

Keywords:
Residential segregation
Structural racism
Racial composition
Health disparities
Pregnancy outcome
Preterm birth
Low birth weight
Infant mortality

ABSTRACT

Rationale: Persistent racial disparities in adverse birth outcomes are not fully explained by individual-level risk factors. Racial residential segregation—degree to which two or more groups live apart from one another—may contribute to the etiology of these birth outcome disparities. Our aim was to assess associations between segregation and adverse birth outcomes by race. This review focused on formal measures of segregation, using Massey and Denton's framework (1998) that identifies five distinct operationalizations of segregation, in addition to proxy measures of segregation such as racial composition, in order to gain a deeper understanding of the operationalizations of segregation most salient for birth outcomes.

Method: Review and meta-analyses were conducted using PubMed, PsycINFO and Web of Science and included articles from inception through April 30, 2017.

Results: Forty-two articles examined associations between segregation and adverse birth outcomes among Black and White mothers separately. Meta-analyses showed that among Black mothers, exposure was associated with increased risk of preterm birth ($OR = 1.17$, 95% $CI = 1.10, 1.26$), and low birth weight ($OR = 1.13$, 95% CI=1.06, 1.21), and Black racial composition was associated with increased risk of preterm birth ($OR = 1.20$, 95% CI=1.05, 1.37), among those living in most- compared to least-segregated neighborhoods. Few studies were conducted among White mothers and only exposure was associated with increased risk of preterm birth and low birth weight. Qualitative analyses indicated that among Black mothers, exposure and hypersegregation were associated with multiple adverse birth outcomes; findings were mixed for evenness and clustering.

Conclusions and future directions: Associations between segregation and adverse birth outcomes differ by race. Methodological heterogeneity between studies may obscure true associations. Research can be advanced through use of multilevel frameworks and by examining mechanistic pathways between segregation and adverse birth outcomes. Elucidation of pathways may provide opportunities to intervene to reduce seemingly intractable racial disparities in adverse birth outcomes.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

Substantial racial disparities in adverse birth outcomes persist in the United States. Preterm birth, a birth occurring before 37 weeks gestation, is one of the most important contributors to infant survival and subsequent health (Mathews et al., 2015). In 2015, preterm birth was 50% higher among non-Hispanic Black (hereafter Black) mothers than non-Hispanic White (hereafter White) mothers (Martin et al., 2017). Low birth weight (<2500 g), another key predictor of infant health, was almost twice as prevalent among Black than among White mothers (13.2% versus 6.9%), and infant mortality (death during first year of life), was 11.1/1000 live births for infants of Black mothers compared to 5.1/1000 for infants of White mothers (Martin et al., 2017; Mathews et al., 2015). Etiology of these disparities is not fully understood. Even after accounting for maternal age, socioeconomic status, health behaviors, and prenatal care, significant differences persist (Berg et al., 2001; El-Sayed et al., 2015), setting the stage for life-long disparities in morbidity and mortality. A growing body of research supports the idea that racism, hierarchical stratification of populations into racial groups and allocation of greater resources and opportunities to those in privileged positions (Bonilla-Silva, 1996), contributes to

* Corresponding author. Yale School of Public Health, 60 College Street, P.O. Box 208034, New Haven, CT 06520-8034, USA.
E-mail address: renee.mehra@yale.edu (R. Mehra).

http://dx.doi.org/10.1016/j.socscimed.2017.09.018
0277-9536/© 2017 Elsevier Ltd. All rights reserved.

racial disparities in disease (Williams and Mohammed, 2013).

Krieger's ecosocial theory of racial inequalities in health promotes examination of racism operating at multiple levels, and informs this review of structural racism and racial disparities in birth outcomes (Krieger, 2012). Krieger argues that multiple socially-patterned exposure pathways operate at individual and structural levels across the life course and generations to harm health. Structurally, determinants of racial inequality include laws, political infrastructure, and neighborhood conditions that disadvantage members of racial minority groups. Williams and Collins (2001) argue that racial residential segregation (hereafter segregation), imposed at the structural level, is a fundamental cause of racial disparities in health. Segregation produces large racial differences in socioeconomic status and mobility in addition to poor-quality social and physical environments that adversely affect health. In addition to the effects of material conditions on health, psychosocially mediated health effects of deprivation are also evident (Marmot and Wilkinson, 2001). For example, substandard housing contributes to psychosocial stress and allostatic load (Krieger and Higgins, 2002), which have been linked to low birth weight (Hilmert et al., 2014), shorter cervical length (Dijkstra et al., 1999), and hormonal disruption among Black pregnant women (Suglia et al., 2010).

Historically, Blacks in northern U.S. cities were segregated from Whites as early as 1880. Across American cities, levels of Black-White segregation were similar to European immigrant-White segregation. However, the unique way Blacks were incorporated into American cities laid the foundation for discriminatory social structures that persist today (for historical overview, see Logan et al., 2015; Massey and Denton, 1993). From 1900 to 1940, American industrialization and the movement of Blacks from southern to northern and rural to urban areas increased segregation. Racial violence in the early twentieth century and institutionalized discrimination in the form of zoning restrictions, racially restrictive covenants of neighborhood associations, federal housing programs, urban renewal legislation, and discriminatory lending and real estate practices reinforced patterns of segregation. By 1940, extreme forms of the ghetto, a predominantly Black zone of a city created by race-based exclusion, had emerged. Black-White segregation peaked mid-century, and since 1970 levels of segregation have fallen modestly as Black migration from the south decelerated then reversed and Black suburbanization began.

Segregation is the degree to which two or more groups live apart from one another. Massey and Denton (1988) identified five distinct operationalizations of segregation: exposure, evenness, clustering, concentration, and centralization. Exposure is the degree to which minority members are exposed to minority or majority members, and is conceptualized as the probability that a minority member shares the same neighborhood as a minority member (isolation) or a majority member (interaction) in a given region. Evenness is the degree to which each neighborhood has the same proportion of minority and majority members as the region as a whole, and is conceptualized as the proportion of a minority group in a given region that would have to change neighborhoods to achieve an even distribution. Clustering is the degree to which minority neighborhoods are contiguous and tightly clustered within a region, creating a single large ghetto or ethnic enclave. Concentration is the degree to which minority members occupy a small proportion of the total area of a region. Centralization is the degree to which a minority group is centrally located within a region (see Fig. 1). Each operationalization is calculated across micro-level spatial units (e.g. neighborhoods) within macro-level spatial units (e.g. regions). Typically, indices used to measure each operationalization range from 0 to 1, with higher scores indicating greater segregation. High levels of segregation on at least four operationalizations is considered hypersegregation (Massey and Denton, 1989).

Operationalizations of segregation are empirically correlated, yet conceptually distinct (Massey and Denton, 1988). For example, exposure and evenness are related but only exposure depends on the relative size of the groups being compared. If minority members constitute a small proportion of the population, exposure to other minority members will be low regardless of the pattern of evenness. Centralized minority groups tend to be concentrated, however concentrated minority groups may be located in central or suburban areas. Clustering is related to concentration and centralization, however regions with the same level of concentration and centralization have different segregation patterns if the minority neighborhoods are clustered together versus separated from one another.

Exposure and evenness are aspatial operationalizations, while clustering, concentration, and centralization are spatial operationalizations whose calculation requires information on location and size of micro-level spatial units (Reardon & O'Sullivan, 2004). Aspatial measures of segregation are susceptible to several problems. The checkerboard problem occurs with a given racial composition, when different spatial rearrangements of micro-level units do not change the measure of segregation at the macro level (White, 1983). The modifiable areal unit problem (MAUP) refers to sensitivity of analytical results to the spatial unit at which data are aggregated (Fotheringham and Wong, 1991). Spatial versions of exposure and evenness have been developed, which may overcome both problems (Anthopolos et al., 2011; Reardon & O'Sullivan, 2004). Reardon and O'Sullivan (2004) contend that there may be two rather than five distinct operationalizations of segregation, namely spatial exposure-isolation and spatial evenness-clustering, with centralization and concentration being subcategories of spatial evenness-clustering. However, as Kramer and Hogue (2009) note, few studies have used spatial measures of segregation and the meaningfulness of these measures for understanding associations between segregation and health is yet to be determined.

White and Borrell (2011) differentiate between formal measures—the aforementioned five operationalizations—and proxy measures such as racial composition, typically operationalized as the proportion of the population in an area that is a racial minority. Although formal and proxy measures of segregation are related conceptually, proxy measures fail to take into account relative location of groups within an area. Spatial versions of proxy measures also have been developed, which may overcome this limitation (Mason et al., 2011a, b; Mason et al., 2010).

Measures of segregation are differentially conceptually relevant to health outcomes. Exposure is relevant to health because it reflects geographic isolation from economic and educational opportunities and resources (Wilson, 1987), and is associated with many neighborhood characteristics (e.g. multiple indicators of housing, neighborhood quality, socioeconomic characteristics, and violence; Denton, 1994; Shihadeh and Flynn, 1996). Evenness has the least clear conceptual relevance for health (Acevedo-Garcia et al., 2003), in part because it is associated with few neighborhood characteristics (Denton, 1994). Clustering may constrain social and economic opportunities for the segregated group, reducing access to social and health services (Acevedo-Garcia et al., 2003). However, clustering may also be associated with health-promoting factors such as social support, cohesion, and political empowerment that lead to improved health outcomes (Bell et al., 2006). Concentration and centralization tend to confine minorities to central city areas that are characterized by old and substandard housing, which may affect physical and mental health (Krieger and Higgins, 2002; Massey and Denton, 1988). Hypersegregation accumulates across five operationalizations simultaneously and is associated with

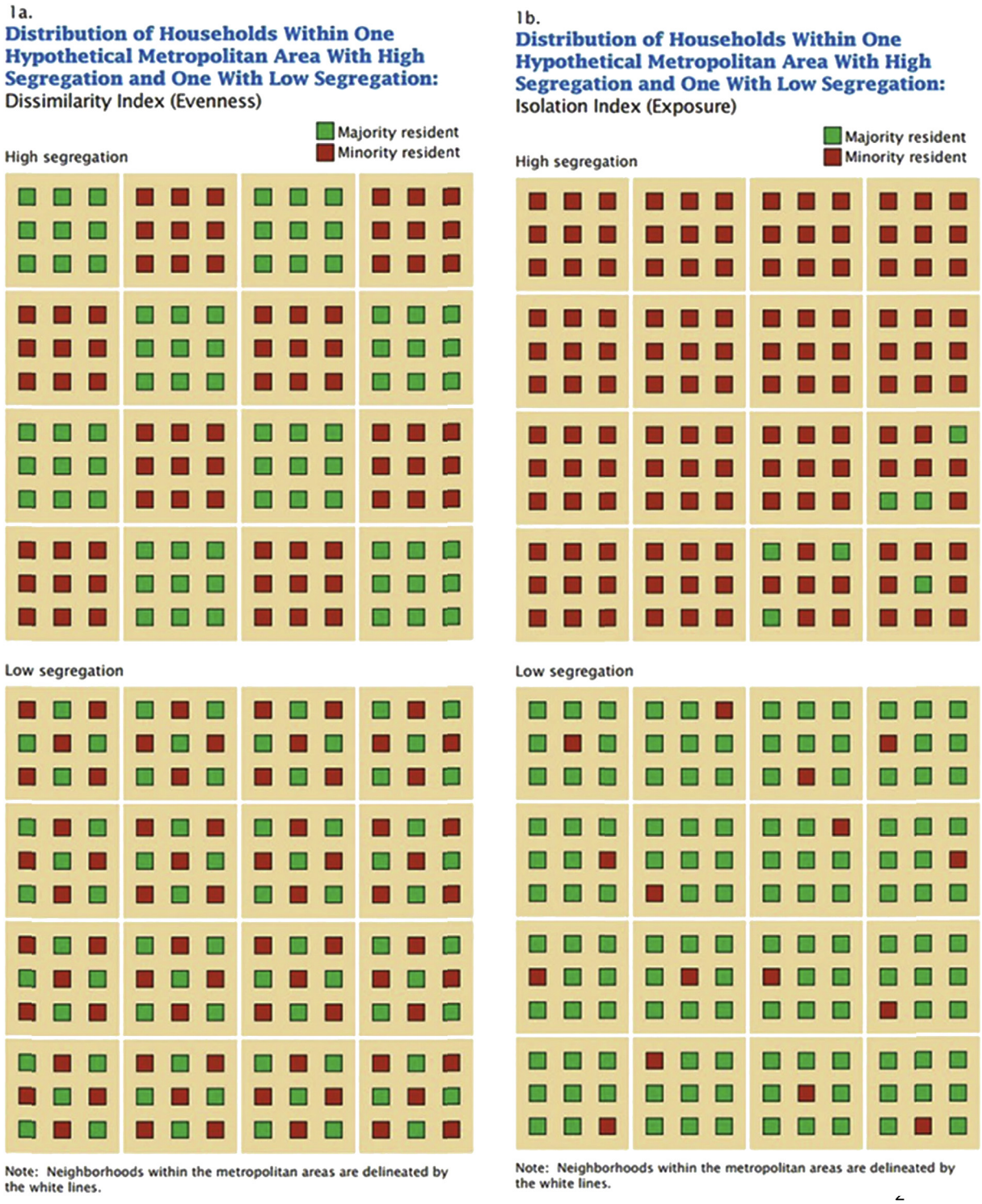

Fig. 1. Distribution of households within a hypothetical metropolitan area for each formal measure of segregation.
Source: Iceland, J., Weinberg, D.H., & Steinmetz, E. (2002). *Racial and ethnic residential segregation in the United States: 1980-2000*. Washington, DC: U.S. Government Printing Office.

multiple measures of social and economic well-being (Denton, 1994; Massey and Denton, 1989). Minority racial composition may be differentially related to health as it is associated with increased levels of deprivation, and psychological advantage that is mediated by reduced exposure to prejudice and increased social support (Becares et al., 2009; Halpern, 1993).

Segregation may be more detrimental to the health of members of racial minority groups than majority groups, as higher segregation is associated with lower educational attainment, employment, and income among Blacks but not Whites (Cutler and Glaeser, 1997). Segregation has been shown to be associated with poorer self-rated health among Blacks but not Whites: specifically, Black

isolation but not White isolation was associated with poorer health among Blacks but not Whites (Subramanian et al., 2005).

This systematic review and meta-analysis synthesize existing evidence on the association between segregation and adverse birth outcomes, including preterm birth, low birth weight, and infant mortality. Previous systematic reviews have examined associations between racial composition and morbidity, mortality and health behaviors, including birth outcomes (Becares et al., 2012); associations between segregation and health outcomes (Kramer and Hogue, 2009); and associations between social determinants, including segregation, and birth outcomes (Kim and Saada, 2013). A recent systematic review and meta-analysis examined the association between neighborhood deprivation (e.g. poverty, deprivation, segregation, crime) and birth outcomes (Ncube et al., 2016), including one study that examined the association between segregation and low birth weight. This review extends previous work by examining both formal and proxy measures of segregation for Black and White mothers separately, and exploring how different operationalizations of segregation are differentially associated with adverse birth outcomes.

As suggested by Krieger's ecosocial framework, racism influences individuals' lives—and ultimately health—via multiple pathways, making a comprehensive examination of multiple measures of segregation vital for understanding the mechanisms underlying complex associations between structural inequality and key health indicators like birth outcomes. Based on previous research (Becares et al., 2012; Kramer and Hogue, 2009), we hypothesize that the association between segregation and birth outcomes will differ across operationalizations of segregation, with higher levels of exposure, evenness, concentration and centralization, but lower levels of clustering, being associated with adverse birth outcomes among Black but not White mothers. Also, we hypothesize that there will be inconsistent evidence for the association between minority racial composition and adverse birth outcomes.

2. Methods

2.1. Search strategy and study selection

We conducted a systematic review of articles published in PubMed, PsycINFO, and Web of Science from inception through April 30, 2017. Search strategy is provided in Supplementary Materials. RM examined articles for relevance based on title and abstract, then full-text articles were assessed for inclusion.

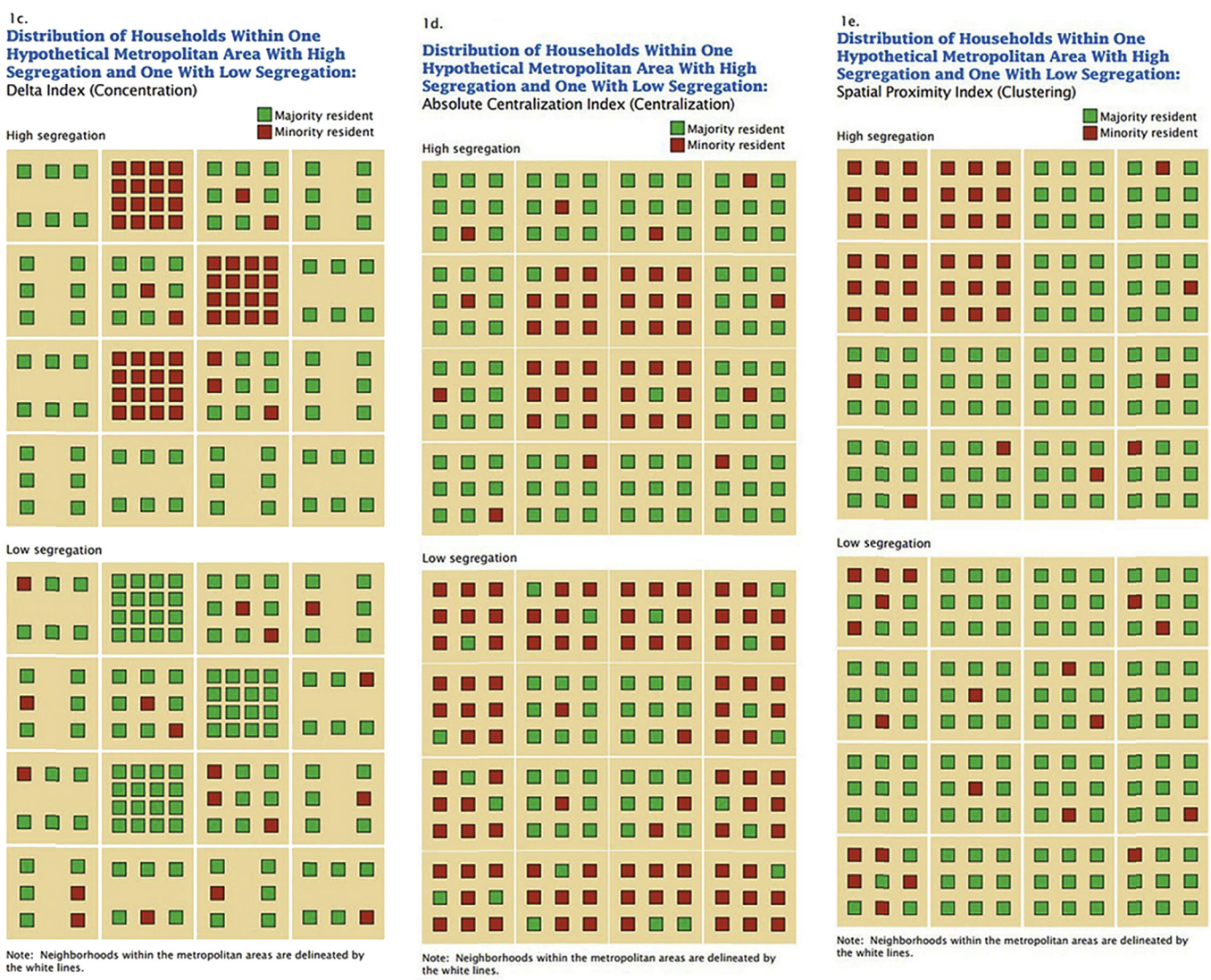

Fig. 1. (*continued*).

Reference lists of screened articles and previous reviews were examined for additional eligible articles. Inclusion criteria were: empirical studies published in peer-reviewed journals; conducted in the U.S. among Black and White mothers separately because of uniquely high levels of segregation experienced by Blacks, paucity of research among other racial/ethnic groups, and differential effect of segregation by race; and assessment of associations between formal or proxy measures of segregation and adverse birth outcomes. Formal measures of segregation included indices of Black-White exposure, evenness, clustering, concentration, centralization, and hypersegregation; proxy measures included Black racial composition. Adverse birth outcomes included but were not limited to preterm birth, low birth weight, and infant mortality (Fig. 2). Studies were not excluded based on language, but only databases of English-language abstracts were searched.

2.2. Data extraction

Data were extracted from eligible studies by RM, and were confirmed independently by LMB and JRI with >99% agreement. Dimensions collected included study methods, population, measures of segregation, birth outcomes, covariates, and results.

2.3. Assessment of bias

Risk of bias in individual studies was assessed using the Quality Assessment Tool from the Effective Public Health Practice Project (2003). Studies were not excluded based on risk of bias; however, methodological limitations were considered in interpretation of results. Risk of bias across studies was evaluated using Cochrane Collaboration's summary assessment (Higgins and Green, 2011).

2.4. Method of synthesis

Effect estimates from fully adjusted models were included. Results for racial composition were not included if studies assessing associations with formal measures of segregation explicitly controlled for racial composition (Bell et al., 2006; Britton and Shin, 2013; LaVeist, 1992; Walton, 2009). Results were presented separately for Black and White mothers. Primary effect measures were risk ratio, odds ratio, risk difference, and standardized or unstandardized beta coefficients.

2.5. Statistical analysis

Meta-analyses were conducted to compare the odds of an adverse birth outcome by race among those in most- compared to least-segregated areas. Pooled effect sizes were calculated for each operationalization of segregation and birth outcome because of the predicted independent association for each comparison. Two or more studies were available to assess associations between segregation measures of exposure and racial composition, and preterm birth and low birth weight. For Black mothers, six of 14 studies on preterm birth and six of 13 studies on low birth weight in this review, contributed data to the meta-analyses. For White mothers, two of six studies each on preterm birth and low birth weight were included in meta-analyses. One study presented data for five separate counties, each included as an independent comparison.

Inverse variance heterogeneity models were run to avoid underestimation of statistical error and spuriously overconfident estimates from random effects models (Doi et al., 2015). Between-study heterogeneity was assessed using the I^2 statistic (Higgins and Thompson, 2002). Publication bias was assessed with the Doi plot and LFK index of asymmetry (Barendregt and Doi). Analyses

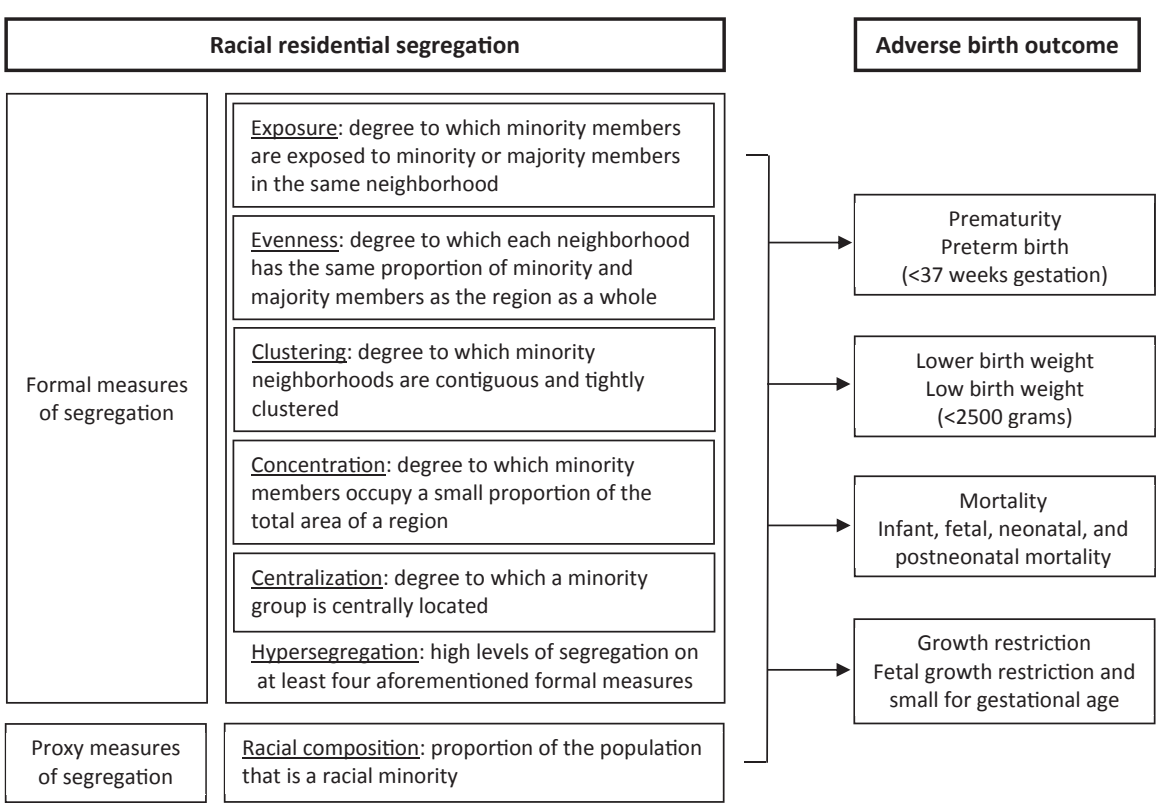

Fig. 2. Effects of formal and proxy measures of racial residential segregation on adverse birth outcomes assessed in this systematic review.

were conducted using MetaXL (version 5.3, EpiGear International Pty Ltd).

3. Results

3.1. Description of included studies

A total of 1651 articles were identified (k = 168 duplicates excluded). Eighty-three full text articles were reviewed; 35 met inclusion criteria; seven additional articles were identified from published studies. In total, 42 studies were included in the review, and eight studies were included in meta-analyses (Fig. 3). Table 1 summarizes study characteristics. Studies may have examined more than one measure of segregation and/or adverse birth outcome using more than one spatial unit. Most studies (k = 25) examined associations with formal measures of segregation and the most common birth outcome was preterm birth (k = 14). Fifteen studies were cross-sectional ecologic studies, two were time-series ecologic studies, 22 were multilevel analyses (seven were cluster-level corrected), two were cross-sectional non-multilevel analyses, and one cohort study used non-multilevel analysis. Source of study populations included vital statistics (k = 35), cohorts of women or infants (k = 4), and hospital database (k = 1). The first study to examine associations between segregation and birth outcomes was published in 1950 (Yankauer, 1950).

3.2. Assessment of bias

Quality of individual studies ranged from moderate to strong. Weakest components across studies were selection bias, study design, and confounders. Bias may occur due to non-representative samples, non-independent assessment of exposures and outcomes, and inadequate control for confounders due to reliance on vital statistics. Therefore, results should be interpreted with caution. Asymmetry in meta-analytic effects (LFK: 1.91 to 4.51) may indicate publication bias, absence of small studies with null findings, or other selection biases.

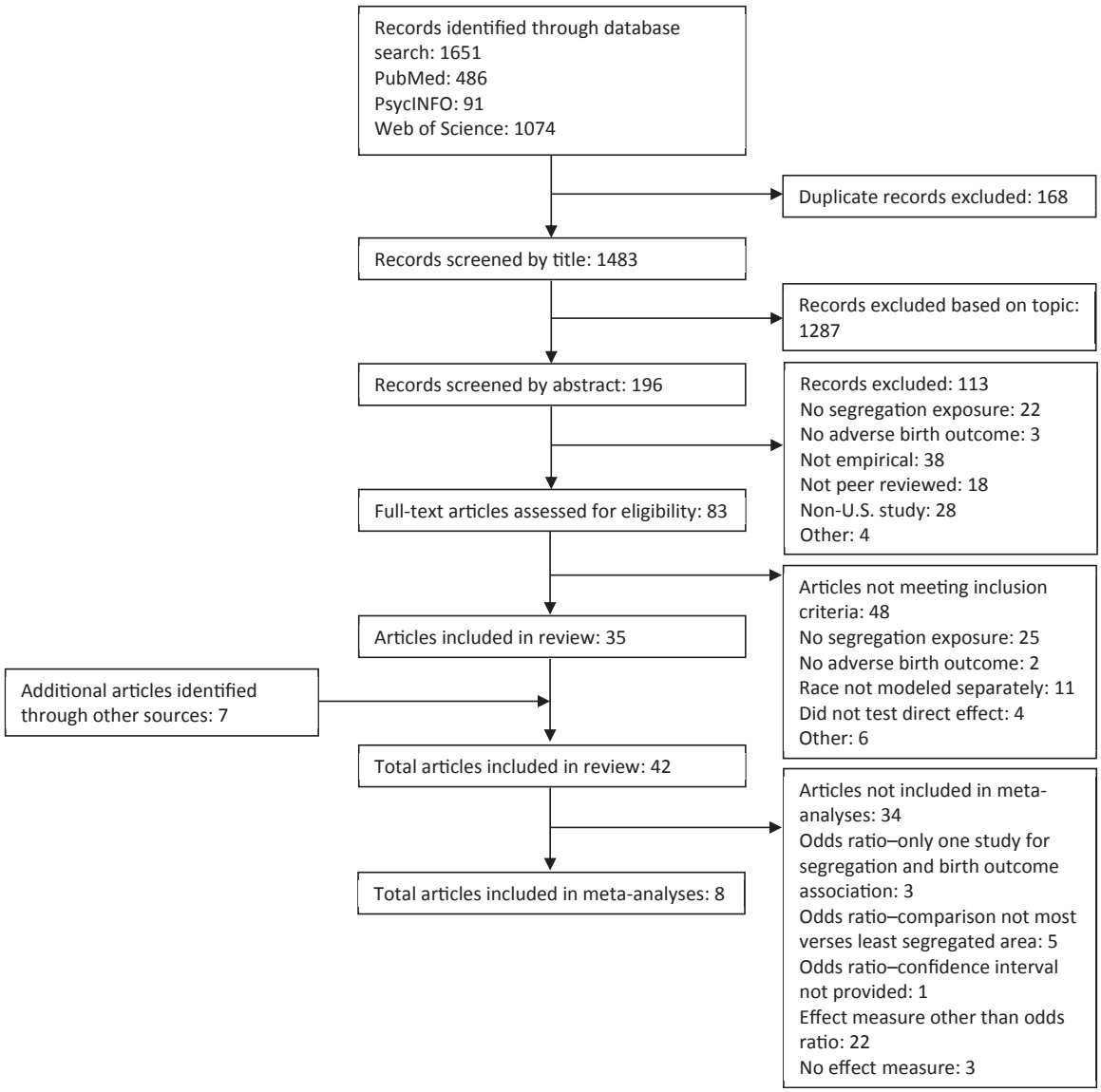

Fig. 3. Flowchart of selection of articles.

Table 1
Summary of included studies ($k = 42$).

Study attributes	Number of studies
Racial residential segregation	
Formal measures	25
Exposure	13
Evenness	15
Clustering	4
Concentration	1
Centralization	2
Other	1
Hypersegregation	2
Proxy measures (black racial composition)	19
Adverse birth outcome	
Prematurity	18
Preterm birth	14
Very preterm birth	5
Moderate preterm birth	2
Lower birth weight	17
Low birth weight	13
Lower birth weight (grams)	7
Mortality	15
Infant	11
Fetal	3
Neonatal	2
Postneonatal	5
Growth restriction	2
Fetal growth restriction	1
Small for gestational age	1
Spatial unit (macro level)	
Census block or block group	4
Census tract (single or aggregated)	14
City/urban place	5
ZIP Code	1
Metropolitan statistical area	11
County/Public Use Metadata Area	5
State	2
Other	2

Note. Studies may have examined more than one measure of segregation, adverse birth outcome, or spatial unit.

3.3. Association between segregation and adverse birth outcomes

Detailed description of studies is included as Supplementary Material. Results can be summarized as follows: (1) meta-analyses showed that exposure was associated with increased risk of adverse birth outcomes among Black and White mothers, and racial composition was associated with increased risk of preterm birth among Black mothers; (2) few studies examined each measure of segregation with each birth outcome, particularly among White mothers; (3) heterogeneity in associations is related to study design and potential moderators; and (4) differential associations by race occur such that segregation was generally associated with increased risk of adverse birth outcomes among Black mothers, and with decreased risk or no risk among White mothers.

For the meta-analyses, the odds of preterm birth and low birth weight were higher among mothers living in most- compared to least-segregated neighborhoods, as measured by exposure (Figs. 4 and 5). Heterogeneity was high overall (I^2: 50%–80%), however within racial subgroups it was low (I^2: 0%-28%), justifying reporting results separately by race. Among Black mothers, odds of preterm birth were 17% higher ($OR = 1.17$, 95% $CI = 1.10, 1.26$), and odds of low birth weight were 15% higher ($OR = 1.13$, 95% $CI = 1.06, 1.21$) among mothers living in most- compared to least-segregated neighborhoods. One study that included five separate counties showed that among White mothers, the magnitude of the association between exposure and preterm birth ($OR = 1.53$, 95% $CI = 1.33, 1.77$), and exposure and low birth weight ($OR = 1.98$, 95% $CI = 1.70, 2.31$) was higher relative to Black mothers. However, caution should be used when interpreting data from a single study. In contrast, Black racial composition was associated with increased risk of preterm birth among Black ($OR = 1.20$, 95% $CI = 1.05, 1.37$), but not White mothers ($OR = 1.12$, 95% $CI = 0.96, 1.29$) (Fig. 6). There was no association between Black racial composition and low birth weight among Black or White mothers (Fig. 7). Heterogeneity in associations between Black racial composition and preterm birth ($I^2 = 63\%$) and low birth weight ($I^2 = 76\%$) among Black mothers may be explained by cut-off values used for the most- and least-segregated neighborhoods. Specifically, studies with higher cut-off values and the largest differentiation between most- and least-segregated neighborhoods had higher magnitudes of association.

Given the heterogeneity in primary effect measures across the 42 studies in this review, we performed a qualitative assessment of broader patterns across all studies and present insights into associations and possible sources of heterogeneity (Table 2). Among Black mothers, all formal measures of segregation were examined in relation to one or more birth outcomes, although in some categories the number of studies was very small. There is a paucity of studies that included White mothers: none examined associations with clustering, concentration, or hypersegregation. Nonetheless, some conclusions may be drawn by synthesizing findings across studies.

Among Black mothers, segregation was associated with increased risk of an array of adverse birth outcomes. In categories with the greatest number of studies, we observed that exposure was associated with increased risk of preterm birth and low birth weight; evenness was associated with increased risk of fetal, postneonatal, and infant mortality; and racial composition was associated with increased risk of preterm birth and infant mortality among subgroups of Blacks. However, findings from studies on mortality tended to be of poorer quality and should be interpreted cautiously. Although the number of studies was smaller, clustering was associated with increased risk of low birth weight, and hypersegregation was associated with increased risk of multiple adverse birth outcomes among Black mothers. Each study using spatial measures of segregation found at least one association with increased risk of adverse birth outcomes, but the same was not found in studies using aspatial measures. Herein, we provide a more detailed assessment disaggregated by operationalization of segregation.

3.4. Formal measures of segregation

Exposure was commonly measured with the isolation index, which measures the extent to which members of the minority group are exposed only to one another, rather than to members of the majority group (Massey and Denton, 1988). Among Black mothers, exposure was associated with increased risk of preterm birth, very preterm birth (<32 weeks gestation), moderate preterm birth (32–37 weeks gestation), low birth weight, lower birth weight (grams), and fetal growth restriction. Among White mothers, exposure was associated with increased risk of low birth weight and lower birth weight (grams). Exposure was not associated with infant mortality among Black or White mothers.

Evenness was commonly measured with the index of dissimilarity, which conceptually refers to the proportion of minority residents required to change neighborhoods to obtain an even distribution of minorities across a region (James and Taueber, 1985; Massey and Denton, 1988). In general, evenness was not associated with preterm birth or low birth weight among Black or White mothers. However, in earlier ecological studies with weak adjustment for confounders, evenness was associated with increased risk

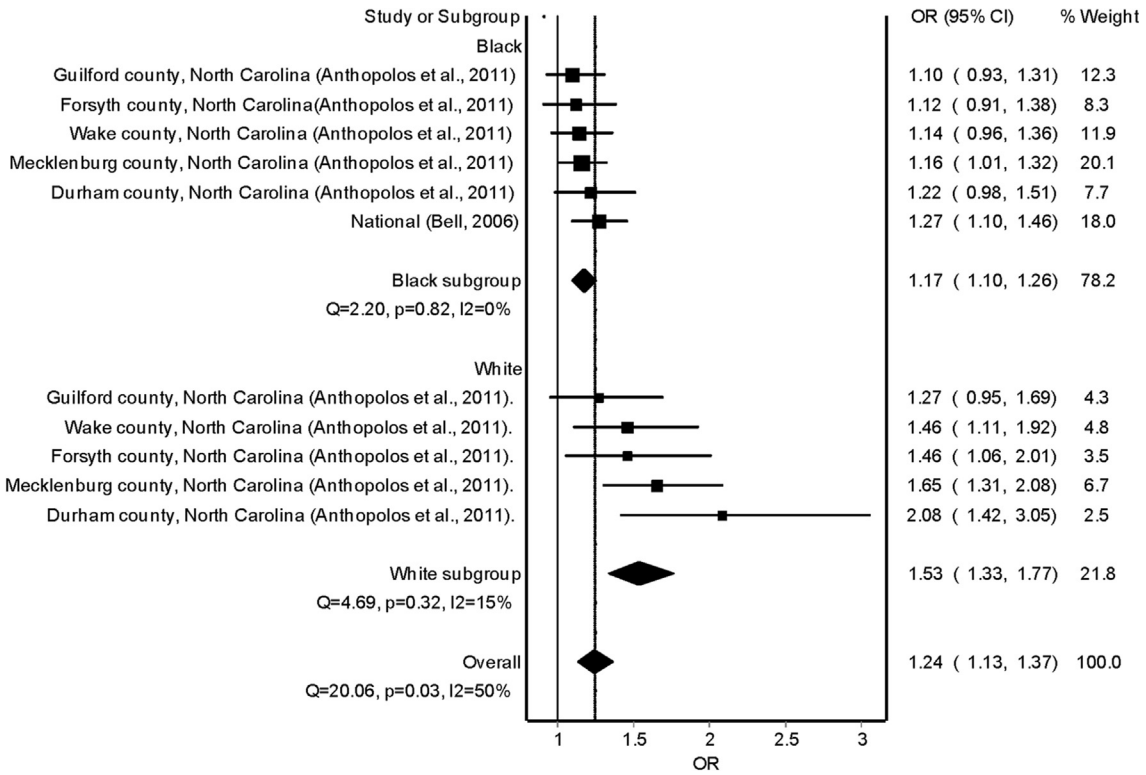

Fig. 4. Odds ratios and confidence intervals of association between exposure and preterm birth.

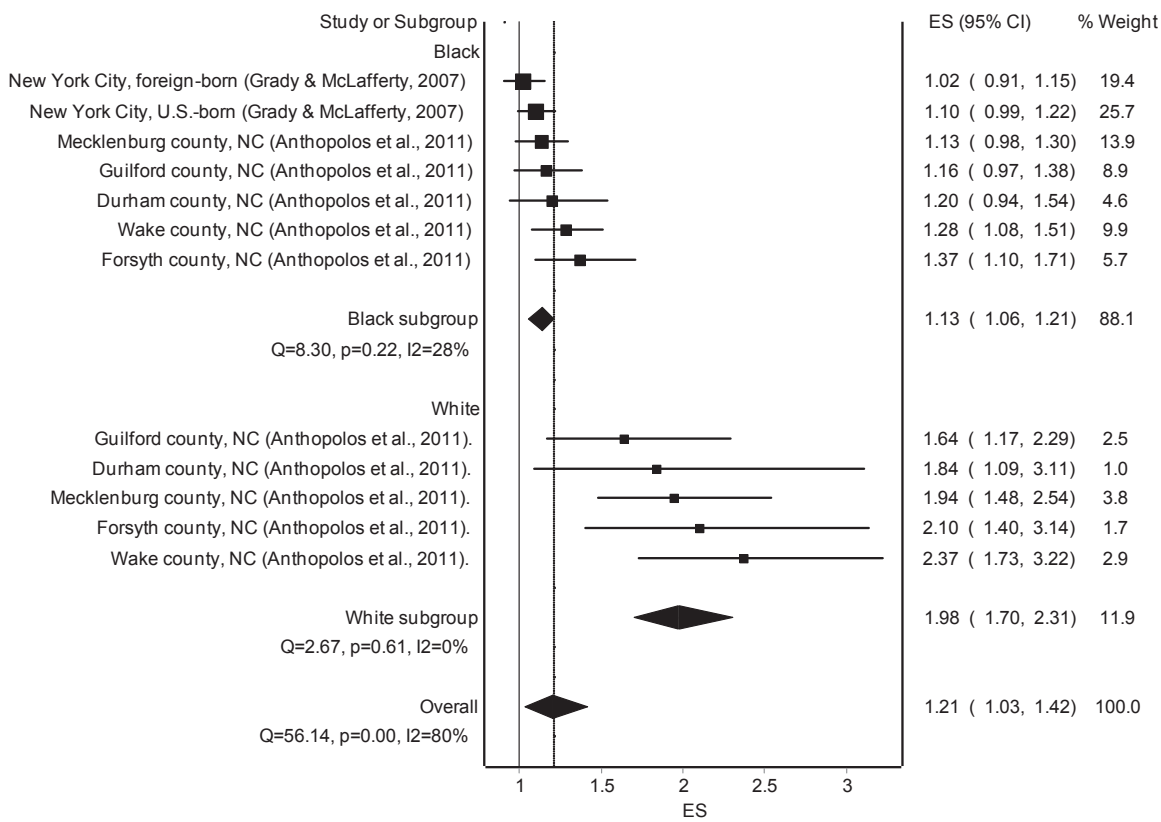

Fig. 5. Odds ratios and confidence intervals of association between exposure and low birth weight.

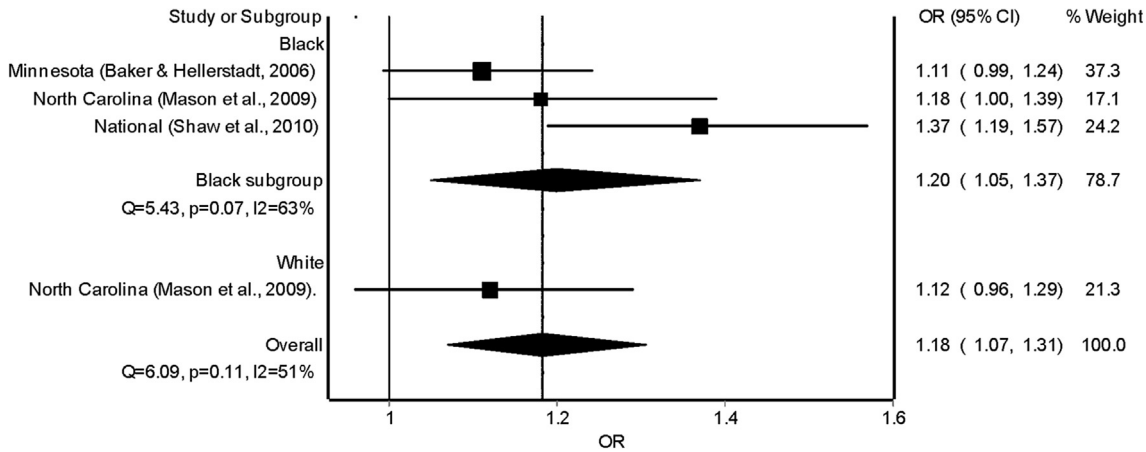

Fig. 6. Odds ratios and confidence intervals of association between Black racial composition and preterm birth.

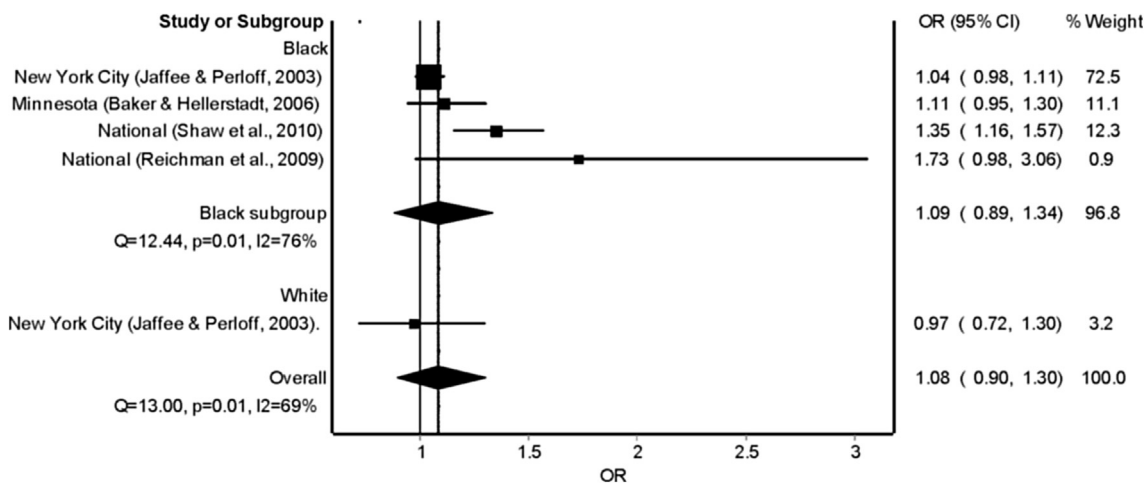

Fig. 7. Odds ratios and confidence intervals of association between Black racial composition and low birth weight.

of infant mortality, fetal mortality (deaths ≥20 weeks gestation/1000 live births plus fetal deaths), and postneonatal mortality (deaths 28–365 days of age/1000 live births) among Black mothers. Findings among White mothers were less consistent, with studies finding increased, decreased, or no risk of adverse birth outcomes.

Among Black mothers, clustering was associated with increased risk of preterm birth and low birth weight. However, results are equivocal with other studies showing that clustering was associated with decreased risk of very preterm birth and lower birth weight (grams). There were no associations between concentration or centralization and adverse birth outcomes. No associations were found between clustering and infant mortality or fetal growth restriction. In one study, the Gorard index of segregation (Gorard, 2000), related to measures of evenness and concentration, was associated with decreased risk of low birth weight among Black mothers only. Among Black mothers, evidence is suggestive of an association between hypersegregation and increased risk of multiple adverse birth outcomes.

3.5. Proxy measure of segregation: racial composition

Among Black mothers, Black racial composition was associated with increased risk of preterm birth. However, results were equivocal, with some studies finding no risk. In general, Black racial composition was not associated with birth weight or small for gestational age. In one study, White composition was not associated with birth weight (Reichman et al., 2009). Overall among White mothers, racial composition was not associated with preterm birth, low birth weight, or small for gestational age. In earlier ecological studies with weak adjustment for confounders, racial composition was associated with increased risk of infant, fetal, neonatal and postneonatal mortality among Black and White mothers. Meanwhile, a more recent multilevel analysis found no association between racial composition and infant mortality among Black mothers (Shaw et al., 2010).

3.6. Differential associations by race

Twenty-one of the 43 segregation-birth outcome comparisons in the 26 studies that included both Black and White mothers found the same direction of association for both groups. Thirteen comparisons found no association between segregation and adverse birth outcomes for Black or White mothers. Eight comparisons that found an increased risk of adverse birth outcomes among both Black and White mothers were primarily from poor-quality ecologic studies on mortality. A common pattern among the 22 comparisons with discordant results by race was increased risk of adverse birth outcomes among Black mothers, and decreased risk (Brown et al., 2012; LaVeist, 1989, 1990, 1992, 1993; Mason et al., 2011b), or no risk (Bird, 1995; McFarland and Smith, 2011; Pickett

Table 2
Associations between racial residential segregation and adverse birth outcomes by race in 42 included studies.

Racial Residential Segregation	Adverse Birth Outcome	Black		White	
		Studies observing at least one association with increased risk of adverse birth outcomes	Studies observing no associations with increased risk of adverse birth outcomes	Studies observing at least one association with increased risk of adverse birth outcomes	Studies observing no associations with increased risk of adverse birth outcomes
Formal measures					
Exposure	Prematurity[a]	6 studies Britton and Shin, 2013; Nyarko and Wehby, 2012; Anthopolos et al., 2011; Kramer et al., 2010; Kramer and Hogue, 2008; Bell et al., 2006	1 study Mendez et al., 2014	1 study Anthopolos et al., 2011	1 study Mendez et al., 2014
	Lower birth weight[b]	7 studies Nyarko & Wehby, 2012; Anthopolos et al., 2011; McFarland and Smith, 2011; Walton, 2009; Grady and Ramirez, 2008; Grady and McLafferty, 2007; Bell et al., 2006	–	2 studies Anthopolos et al., 2011; Grady and Ramirez, 2008	1 study McFarland and Smith, 2011
	Mortality[c]	–	3 studies McFarland and Smith, 2011; Hearst et al., 2008; Guest et al., 1998 ǂ	–	2 studies McFarland and Smith, 2011; Guest et al., 1998 ǂ
	Growth restriction[d]	1 study Bell et al., 2006	–	–	–
Evenness	Prematurity[a]	1 study Nyarko and Wehby, 2012	3 studies Mendez et al., 2014; Britton and Shin, 2013; Kramer and Hogue, 2008	–	1 study Mendez et al., 2014
	Lower birth weight[b]	1 study Nyarko and Wehby, 2012	3 studies Austin et al., 2016 ǂ; McFarland and Smith, 2011; Ellen, 2000	–	3 studies Austin et al., 2016 ǂ; McFarland and Smith, 2011; Ellen, 2000
	Mortality[c]	7 studies Brown et al., 2012; Bird, 1995; LaVeist, 1993, 1992, 1990, 1989; Polednak, 1991 ǂ	2 studies McFarland and Smith, 2011; Polednak, 1996 ǂ	1 study McFarland and Smith, 2011	8 studies Brown et al., 2012[e]; Bird 1995; LaVeist, 1993[e], 1992[e] 1990[e] 1989[e]; Polednak 1996 ǂ, 1991 ǂ
Clustering	Prematurity[a]	1 study Bell et al., 2006	1 study Kramer et al., 2010	–	–
	Lower birth weight[b]	2 studies McFarland and Smith, 2011; Walton, 2009	1 study Bell et al., 2006	–	–
	Mortality[c]	–	1 study McFarland and Smith, 2011	–	–
	Growth restriction[d]	–	1 study Bell et al., 2006	–	–
Concentration	Lower birth weight[b]	–	1 study McFarland and Smith, 2011	–	–
	Mortality[c]	–	1 study McFarland and Smith, 2011	–	–
Centralization	Lower birth weight[b]	–	2 studies McFarland and Smith, 2011; Ellen, 2000	–	1 study Ellen, 2000
	Mortality[c]	–	1 study McFarland and Smith, 2011	–	–
Other	Lower birth weight[b]	–	1 study Herd et al., 2015[e]	–	1 study Herd et al., 2015
Hypersegregation	Prematurity[a]	1 study Osypuk and Acevedo-Garcia, 2008	–	–	–
	Lower birth weight[b]	1 study McFarland and Smith, 2011	–	–	–
	Mortality[c]	1 study McFarland and Smith, 2011	–	–	–
Proxy measures					
Black racial composition	Prematurity[a]	8 studies Carmichael et al., 2014; Mason et al., 2011a, 2011b, 2010, 2009; Shaw et al., 2010; Baker and Hellerstedt, 2006; Pickett et al., 2002	3 studies Mendez et al., 2014; Phillips et al., 2009; Masi et al., 2007	2 studies Mason et al., 2009; Masi et al., 2007	3 studies Mendez et al., 2014; Mason et al., 2011b[e]; Pickett et al., 2002
	Lower birth weight[b]	2 studies Shaw et al., 2010; Reichman et al., 2009	5 studies Madkour et al., 2014[e]; Masi et al. 2007; Baker & Hellerstedt, 2006; Jaffee and Perloff, 2003; Buka et al., 2003	1 study Masi et al., 2007	3 studies Madkour et al., 2014; Jaffee and Perloff, 2003; Buka et al., 2003
	Mortality[c]				

Table 2 (*continued*)

Racial Residential Segregation	Adverse Birth Outcome	Black		White	
		Studies observing at least one association with increased risk of adverse birth outcomes	Studies observing no associations with increased risk of adverse birth outcomes	Studies observing at least one association with increased risk of adverse birth outcomes	Studies observing no associations with increased risk of adverse birth outcomes
		3 studies Bird, 1995; Yankauer and Allaway, 1958 ǂ; Yankauer, 1950 ǂ	2 studies Shaw et al., 2010; McLeod et al., 2004	2 studies Yankauer and Allaway, 1958 ǂ; Yankauer, 1950 ǂ	2 studies McLeod et al., 2004; Bird, 1995
	Growth restriction[d]	–	1 study Masi et al., 2007	1 study Masi et al., 2007	–

Note. Studies may have examined more than one adverse birth outcome per category (i.e., low birth weight and lower birth weight (grams)) for each measure of segregation. Global rating for study using Effective Public Health Practice Project Quality Assessment Tool; components include selection bias, study design, confounding, blinding, data collection methods, and withdrawals/dropouts: no symbol indicates strong global rating (no components with a weak rating); ǂ indicates moderate global rating (one component with a weak rating).

[a] Prematurity includes preterm birth, very preterm birth and moderate preterm birth.

[b] Lower birth weight includes low birth weight and lower birth weight (grams).

[c] Mortality includes fetal, neonatal, postneonatal, and infant mortality.

[d] Growth restriction includes fetal growth restriction and small for gestational age.

[e] Studies observing at least one association where segregation was associated with decreased risk of adverse birth outcomes.

et al., 2002; Polednak, 1991) among White mothers. Less common were studies with decreased risk among Black but not White mothers (Herd et al., 2015; Madkour et al., 2014).

4. Discussion

This systematic review and meta-analysis synthesizes the literature on segregation and adverse birth outcomes in 42 studies over more than 65 years. When disaggregating by operationalization of segregation and birth outcome among Black and White mothers, the number of studies in many categories is small. Although we note heterogeneous associations, certain operationalizations of segregation were associated with increased risk of adverse birth outcomes, particularly among Black mothers.

Exposure and hypersegregation were associated with increased risk of numerous adverse birth outcomes among Black mothers. We expected exposure and hypersegregation to be associated with birth outcomes, as they are associated with multiple indicators of social and economic well-being. Evenness is the most widely used measure of segregation but has the least clear conceptual relevance for health, which may explain why it was not consistently associated with adverse birth outcomes. Clustering was expected to have differential associations with health as it is related to deleterious socioeconomic and physical environments, and advantageous social environments, and was associated with improved birth outcomes in some studies. Few studies examined associations between concentration or centralization and adverse birth outcomes.

Inconsistent evidence for the association between racial composition and adverse birth outcomes may depend on context, given that minority racial composition is associated with both increased levels of material deprivation and improved psychosocial health (Becares et al., 2009; Halpern, 1993). For example, Black racial composition is associated with poorer birth outcomes among U.S.-born, but not foreign-born, Black mothers and these associations may be modified by neighborhood deprivation. Alternatively, equivocal associations may be related to methodological heterogeneity in study design (ecologic versus individual), and adjustment for confounding.

Studies of White mothers were limited, and generally neither formal nor proxy measures of segregation were associated with poorer birth outcomes. Whites may not experience poorer health outcomes in segregated regions if they live in neighborhoods that are of higher quality both physically and economically (Wilson, 1987).

Local spatial measures of exposure were associated with increased risk of adverse birth outcomes among both Black and White mothers in multilevel analyses (Anthopolos et al., 2011; Grady and Ramirez, 2008). This may reflect smaller spatial units yielding higher levels of segregation, or mediation through deleterious social and physical environments regardless of race. Further research is needed on the role of spatial measures on adverse birth outcomes. Additionally, research on the role of other racial/ethnic-specific segregation on birth outcomes is limited.

4.1. Limitations

A small number of studies in this systematic review were included in the meta-analyses and effect modification was unable to be assessed due to study heterogeneity. This review was restricted to Black and White populations and to studies conducted in the U.S. Publication bias may have resulted in exclusion of studies with null findings. Although this review included only peer-reviewed studies, in an examination of abstracts from dissertations that otherwise met inclusion criteria, three of four were published and findings were included in this review (Hearst et al., 2008; Mason et al., 2010, 2011a; Mendez et al., 2014), two of which reported null findings.

Methodologic limitations of the studies included heterogeneity in study design and analysis, source population, measure of segregation, spatial unit of analysis, specification of segregation, adjustment for potential confounders or covariates on the causal pathway, and assessment of moderation. In earlier ecologic studies with weak adjustment for confounding, segregation was associated with higher aggregate-level adverse birth outcomes. However, more recent multilevel analyses have produced mixed results at the individual level. Multilevel analyses can identify independent associations between contextual variables and birth outcomes, and how contextual variables relate to each other and combine with individual characteristics in additive and interactive ways to influence health (Williams and Collins, 2001). Most studies were cross-sectional, limiting causal inferences, and only assessed associations with static measures of segregation. A life course approach that includes cumulative effects (Kramer et al., 2014) would advance the research agenda.

Formal and proxy measures are used interchangeably, however formal measures better reflect social processes of racial inequality and represent potential interaction patterns between racial groups (White and Borrell, 2011). Formal and proxy measures are poorly correlated, thus inconsistent conceptualization of segregation may obscure true associations.

Massey and Denton (1988) recommended measuring the multidimensional phenomenon of segregation using several indices. Of 25 studies examining associations with formal measures of segregation, only one examined all five operationalizations, and only eight examined more than one operationalization. Exposure was the only operationalization significantly associated with increased risk of low birth weight and infant mortality when all five operationalizations were taken into account (McFarland and Smith, 2011). Research on other outcomes, such as mortality, found similar results (Yang and Matthews, 2015). After taking clustering into account, exposure was associated with increased risk of moderate and very preterm birth (Kramer et al., 2010), and low birth weight (Walton, 2009). After taking evenness into account, exposure was associated with increased risk of very preterm birth (Kramer and Hogue, 2008). Overall, few studies examined associations between clustering, concentration, centralization, or hypersegregation, and birth outcomes.

Choice of spatial unit affects size and variability of an index, with smaller spatial units generally yielding higher indices of segregation (Massey and Denton, 1988). For formal measures of segregation, macro-level spatial units ranged from aggregated census tracts to MSAs, while micro-level spatial units ranged from census blocks to census tracts. For proxy measures of segregation, spatial units ranged from census blocks to states. Use of continuous or categorical levels of segregation was inconsistent across studies.

Kramer and Hogue (2009) delineate four hypothesized interconnected mechanisms between segregation and health: individual socioeconomic status, neighborhood socioeconomic environment, social capital, and individual risk behaviors and exposure to stressful stimuli. Additionally, chronic diseases partially mediate the association between segregation and low birth weight (Grady and Ramirez, 2008). Therefore, controlling for these mediators may lead to underestimation of the magnitude of association between segregation and adverse birth outcomes. Indeed, controlling for individual-level (Anthopolos et al., 2011; Britton and Shin, 2013; Ellen, 2000; Mason et al., 2009; Reichman et al., 2009) and area-level (Ellen, 2000; Reichman et al., 2009) socioeconomic covariates attenuated this association. Nonetheless, maternal socioeconomic factors may be an important source of confounding as mothers are not randomly distributed across neighborhoods and may self-select into neighborhoods based on socioeconomic characteristics (Anthopolos et al., 2011). Additionally, racial composition, which generally increases as formal measures of segregation increase, may be a source of confounding and should be adjusted for in analyses (White and Borrell, 2011).

Potential moderators were also a source of heterogeneity. Exposure was associated with increased risk of low birth weight among U.S.-born but not foreign-born Black mothers (Grady and McLafferty, 2007). Racial composition was associated with increased risk of preterm birth among Black mothers, particularly those residing in deprived neighborhoods (Mason et al., 2011b), and this association differed by nativity (Mason et al., 2010).

4.2. Conclusions and future research

This comprehensive, systematic review on associations between segregation and adverse birth outcomes reveals important patterns and gaps. Among Black mothers, segregation impacts outcomes at birth that contribute to health across the developmental life course. Research can be advanced through use of multilevel frameworks and by examining mechanistic pathways between segregation and adverse birth outcomes. Elucidation of pathways may provide opportunities to intervene to reduce the seemingly intractable racial disparities in adverse birth outcomes.

Identifying biopsychosocial mechanisms by which segregation influences birth outcomes is an important next step for this field. For example, research shows a protective effect of positive income incongruity on preterm birth among Black mothers in areas of high Black racial composition (Pickett et al., 2005; Vinikoor et al., 2008). Both low and high levels of segregation are associated with higher odds of smoking during pregnancy among Black women, and smoking is associated with adverse birth outcomes (Bell et al., 2007). Studies should examine individual- and area-level mediators to explain differences between races, and should stratify to identify high-risk groups. For example, racial composition is associated with increased risk of preterm birth among African- and U.S.-born Black mothers, but not Caribbean-born Black mothers (Mason et al., 2010). Use of measures such as Jim Crow legislation (Krieger et al., 2014) and redlining, financial institution mortgage lending practices that discriminate on the basis of race/ethnicity and socioeconomic status (Mendez et al., 2014), while showing mixed associations with adverse birth outcomes, may provide avenues to broaden this field of research.

Exposure and hypersegregation were associated with increased risk of multiple adverse birth outcomes among Black mothers. Racial composition may be associated with increased risk of preterm birth among Black mothers. Inconsistent findings across other operationalizations of segregation may arise from different operationalizations having distinct mechanistic pathways to health outcomes (White and Borrell, 2011). Studies should distinguish between using formal and proxy measures of segregation, and provide conceptual justification for the operationalization of segregation being assessed by identifying specific mechanisms linking it to the health outcome (Acevedo-Garcia and Lochner, 2003). Future research also should examine multiple measures of segregation concurrently among both races.

In 1968, the Kerner Commission concluded that "our nation is moving toward two societies, one Black, one White—separate and unequal" (National Advisory Commission on Civil Disorders, 1968). Although the number of hypersegregated metropolitan areas declined by half between 1970 and 2010, the degree of segregation in those areas changed little (Massey and Tannen, 2015). Further research is required to better understand how area-level factors contribute to etiologies of racial disparities in adverse birth outcomes and to identify opportunities for clinical, community, and policy interventions.

Acknowledgements

Ms. Mehra's work was supported by the Herman Fellowship at the Yale School of Public Health. The authors would like to thank Shayna D. Cunningham, Fatma M. Shebl, and Urania Magriples for their insightful reviews of the manuscript.

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.socscimed.2017.09.018.

References

Acevedo-Garcia, D., Lochner, K.A., 2003. Residential segregation and health. In: Kawachi, I., Berkman, L.F. (Eds.), Neighborhoods and Health. Oxford University Press, Inc, New York.

Acevedo-Garcia, D., Lochner, K.A., Osypuk, T.L., Subramanian, S.V., 2003. Future directions in residential segregation and health research: a multilevel approach. Am. J. Public Health 93, 215–221.

Anthopolos, R., James, S.A., Gelfand, A.E., Miranda, M.L., 2011. A spatial measure of neighborhood level racial isolation applied to low birthweight, preterm birth, and birthweight in North Carolina. Spat. Spatiotemporal Epidemiol. 2, 235–246.

Austin, N., Harper, S., Strumpf, E., 2016. Does segregation lead to lower birth weight?: an instrumental variable approach. Epidemiology 27, 682–689.

Barendregt, J.J., & Doi, S.A. MetaXL User Guide Version 5.3. EpiGear International Pty Ltd.
Baker, A.N., Hellerstedt, W.L., 2006. Residential racial concentration and birth outcomes by nativity: do neighbors matter? J. Natl. Med. Assoc. 98, 172–180.
Becares, L., Nazroo, J., Stafford, M., 2009. The buffering effects of ethnic density on experienced racism and health. Health Place 15, 670–678.
Becares, L., Shaw, R., Nazroo, J., Stafford, M., Albor, C., Atkin, K., et al., 2012. Ethnic density effects on physical morbidity, mortality, and health behaviors: a systematic review of the literature. Am. J. Public Health 102, e33–66.
Bell, J.F., Zimmerman, F.J., Almgren, G.R., Mayer, J.D., Huebner, C.E., 2006. Birth outcomes among urban African-American women: a multilevel analysis of the role of racial residential segregation. Soc. Sci. Med. 63, 3030–3045.
Bell, J.F., Zimmerman, F.J., Mayer, J.D., Almgren, G.R., Huebner, C.E., 2007. Associations between residential segregation and smoking during pregnancy among urban African-American women. J. Urban Health 84, 372–388.
Berg, C.J., Wilcox, L.S., d'Almada, P.J., 2001. The prevalence of socioeconomic and behavioral characteristics and their impact on very low birth weight in black and white infants in Georgia. Matern. Child. Health J. 5, 75–84.
Bird, S.T., 1995. Separate black and white infant mortality models: differences in the importance of structural variables. Soc. Sci. Med. 41, 1507–1512.
Bonilla-Silva, E., 1996. Rethinking racism: toward a structural interpretation. Am. Sociol. Rev. 62, 465–480.
Britton, M.L., Shin, H., 2013. Metropolitan residential segregation and very preterm birth among African American and Mexican-origin women. Soc. Sci. Med. 98, 37–45.
Brown, R.A., Hogue, C., Kramer, M.R., 2012. Social and economic determinants of stillbirths in Georgia. Int. J. Disabil. Hum. Dev. 11, 369–375.
Buka, S.L., Brennan, R.T., Rich-Edwards, J.W., Raudenbush, S.W., Earls, F., 2003. Neighborhood support and the birth weight of urban infants. Am. J. Epidemiol. 157, 1–8.
Carmichael, S.L., Cullen, M.R., Mayo, J.A., Gould, J.B., Loftus, P., Stevenson, D.K., et al., 2014. Population-level correlates of preterm delivery among black and white women in the U.S. PLoS One 9, e94153.
Cutler, D.M., Glaeser, E.L., 1997. Are ghettos good or bad? Q. J. Econ. 112.
Denton, N.A., 1994. Are african americans still hypersegregated? In: Bullard, R.D., Grigsby, J.E., Lee, C., Feagin, J.R. (Eds.), Residential Apartheid: the American Legacy. CAAS Publications, Los Angeles, pp. 49–81.
Dijkstra, K., Janssen, H.C., Kuczynski, E., Lockwood, C.J., 1999. Cervical length in uncomplicated pregnancy: a study of sociodemographic predictors of cervical changes across gestation. Am. J. Obstet. Gynecol. 180, 639–644.
Doi, S.A., Barendregt, J.J., Khan, S., Thalib, L., Williams, G.M., 2015. Advances in the meta-analysis of heterogeneous clinical trials I: the inverse variance heterogeneity model. Contemp. Clin. Trials 45, 130–138.
Effective Public Health Practice Project, 2003. Quality Assessment Tool for Quantitative Studies. Effective Public Health Practice Project.
El-Sayed, A.M., Finkton Jr., D.W., Paczkowski, M., Keyes, K.M., Galea, S., 2015. Socioeconomic position, health behaviors, and racial disparities in cause-specific infant mortality in Michigan, USA. Prev. Med. 76, 8–13.
Ellen, I., 2000. Is Segregation Bad for Your Health? Brookings-Wharton Papers on Urban Affairs, pp. 203–238.
Fotheringham, A.S., Wong, D.W.S., 1991. The modifiable areal unit problem in multivariate statistical analysis. Environ. Plan. A 23, 1025–1044.
Gorard, S., 2000. Education and Social Justice. University of Wales Press, Cardiff.
Grady, S.C., McLafferty, S., 2007. Segregation, nativity, and health: reproductive health inequalities for immigrant and native-born black women in New York City. Urban Geogr. 28, 377–397.
Grady, S.C., Ramirez, I.J., 2008. Mediating medical risk factors in the residential segregation and low birthweight relationship by race in New York City. Health Place 14, 661–677.
Guest, A.M., Almgren, G., Hussey, J.M., 1998. The ecology of race and socioeconomic distress: infant and working-age mortality in Chicago. Demography 35, 23–34.
Halpern, D., 1993. Minorities and mental health. Soc. Sci. Med. 36, 597–607.
Hearst, M.O., Oakes, J.M., Johnson, P.J., 2008. The effect of racial residential segregation on black infant mortality. Am. J. Epidemiol. 168, 1247–1254.
Herd, D., Gruenewald, P., Remer, L., Guendelman, S., 2015. Community level correlates of low birthweight among African American, Hispanic and white women in California. Matern. Child. Health J. 19, 2251–2260.
Higgins, J.P., Green, S., 2011. Cochrane Handbook for Systematic Reviews of Interventions. The Cochrane Collaboration, Version 5.1.0. .
Higgins, J.P., Thompson, S.G., 2002. Quantifying heterogeneity in a meta-analysis. Stat. Med. 21, 1539–1558.
Hilmert, C.J., Dominguez, T.P., Schetter, C.D., Srinivas, S.K., Glynn, L.M., Hobel, C.J., et al., 2014. Lifetime racism and blood pressure changes during pregnancy: implications for fetal growth. Health Psychol. 33, 43–51.
Jaffee, K.D., Perloff, J.D., 2003. An ecological analysis of racial differences in low birthweight: implications for maternal and child health social work. Health Soc. Work 28, 9–22.
James, D.R., Taueber, K.E., 1985. Measures of segregation. In: Tuma, N. (Ed.), Sociological Methodology 1985. Jossey-Bass, San Francisco.
Kim, D., Saada, A., 2013. The social determinants of infant mortality and birth outcomes in Western developed nations: a cross-country systematic review. Int. J. Environ. Res. Public Health 10, 2296–2335.
Kramer, M.R., Cooper, H.L., Drews-Botsch, C.D., Waller, L.A., Hogue, C.R., 2010. Metropolitan isolation segregation and Black-White disparities in very preterm birth: a test of mediating pathways and variance explained. Soc. Sci. Med. 71, 2108–2116.
Kramer, M.R., Dunlop, A.L., Hogue, C.J., 2014. Measuring women's cumulative neighborhood deprivation exposure using longitudinally linked vital records: a method for life course MCH research. Matern. Child. Health J. 18, 478–487.
Kramer, M.R., Hogue, C.R., 2008. Place matters: variation in the black/white very preterm birth rate across U.S. metropolitan areas, 2002-2004. Public Health Rep. 123, 576–585.
Kramer, M.R., Hogue, C.R., 2009. Is segregation bad for your health? Epidemiol. Rev. 31, 178–194.
Krieger, J., Higgins, D.L., 2002. Housing and health: time again for public health action. Am. J. Public Health 92, 758–768.
Krieger, N., 2012. Methods for the scientific study of discrimination and health: an ecosocial approach. Am. J. Public Health 102, 936–944.
Krieger, N., Chen, J.T., Coull, B.A., Beckfield, J., Kiang, M.V., Waterman, P.D., 2014. Jim Crow and premature mortality among the US black and white population, 1960-2009 an age-period-cohort analysis. Epidemiology 25, 494–504.
LaVeist, T.A., 1989. Linking residential segregation to the infant mortality race disparity. Sociol. Soc. Res. 73.
LaVeist, T.A., 1990. Simulating the effects of poverty on the race disparity in postneonatal mortality. J. Public Health Policy 11, 463–473.
LaVeist, T.A., 1992. The political empowerment and health status of African-Americans: mapping a new territory. Am. J. Sociol. 97, 1080–1095.
LaVeist, T.A., 1993. Segregation, poverty, and empowerment: health consequences for African Americans. Milbank Q. 71, 41–64.
Logan, J.R., Zhang, W., Turner, R., Shertzer, A., 2015. Creating the black ghetto: black residential patterns before and during the great migration. Ann. Am. Acad. Pol. Soc. Sci. 660, 18–35.
Madkour, A.S., Harville, E.W., Xie, Y., 2014. Neighborhood disadvantage, racial concentration and the birthweight of infants born to adolescent mothers. Matern. Child. Health J. 18, 663–671.
Marmot, M., Wilkinson, R.G., 2001. Psychosocial and material pathways in the relation between income and health: a response to Lynch et al. BMJ 322, 1233–1236.
Martin, J.A., Hamilton, B.E., Osterman, M.J., Driscoll, A.K., Mathews, T.J., 2017. Births: final data for 2015. Natl. Vital Stat. Rep. 66 (1).
Masi, C.M., Hawkley, L.C., Piotrowski, Z.H., Pickett, K.E., 2007. Neighborhood economic disadvantage, violent crime, group density, and pregnancy outcomes in a diverse, urban population. Soc. Sci. Med. 65, 2440–2457.
Mason, S.M., Kaufman, J.S., Daniels, J.L., Emch, M.E., Hogan, V.K., Savitz, D.A., 2011a. Black preterm birth risk in nonblack neighborhoods: effects of Hispanic, Asian, and non-Hispanic white ethnic densities. Ann. Epidemiol. 21, 631–638.
Mason, S.M., Kaufman, J.S., Daniels, J.L., Emch, M.E., Hogan, V.K., Savitz, D.A., 2011b. Neighborhood ethnic density and preterm birth across seven ethnic groups in New York City. Health Place 17, 280–288.
Mason, S.M., Kaufman, J.S., Emch, M.E., Hogan, V.K., Savitz, D.A., 2010. Ethnic density and preterm birth in African-, Caribbean-, and US-born non-Hispanic black populations in New York City. Am. J. Epidemiol. 172, 800–808.
Mason, S.M., Messer, L.C., Laraia, B.A., Mendola, P., 2009. Segregation and preterm birth: the effects of neighborhood racial composition in North Carolina. Health Place 15, 1–9.
Massey, D.S., Denton, N.A., 1988. The dimensions of residential segregation. Soc. Forces 67, 281–315.
Massey, D.S., Denton, N.A., 1989. Hypersegregation in U.S. metropolitan areas: black and Hispanic segregation along five dimensions. Demography 26, 373–391.
Massey, D.S., Denton, N.A., 1993. American Apartheid: Segregation and the Making of the Underclass. Harvard University Press, Cambridge, MA.
Massey, D.S., Tannen, J., 2015. A research note on trends in black hypersegregation. Demography 52, 1025–1034.
Mathews, T.J., MacDorman, M.F., Thoma, M.E., 2015. Infant mortality statistics from the 2013 period linked birth/infant death data set. Natl. Vital Stat. Rep. 64 (9).
McFarland, M., Smith, C.A., 2011. Segregation, race, and infant well-being. Popul. Res. Policy Rev. 30, 467–493.
McLeod, J.D., Nonnemaker, J.M., Call, K.T., 2004. Income inequality, race, and child well-being: an aggregate analysis in the 50 United States. J. Health & Soc. Behav. 45, 249–264.
Mendez, D.D., Hogan, V.K., Culhane, J.F., 2014. Institutional racism, neighborhood factors, stress, and preterm birth. Ethn. Health 19, 479–499.
National Advisory Commission on Civil Disorders, 1968. Report of the National Advisory Commission on Civil Disorders. Bantam Books, New York.
Ncube, C.N., Enquobahrie, D.A., Albert, S.M., Herrick, A.L., Burke, J.G., 2016. Association of neighborhood context with offspring risk of preterm birth and low birthweight: a systematic review and meta-analysis of population-based studies. Soc. Sci. Med. 153, 156–164.
Nyarko, K.A., Wehby, G.L., 2012. Residential segregation and the health of African-American infants: does the effect vary by prevalence? Matern. Child. Health J. 16, 1491–1499.
Osypuk, T.L., Acevedo-Garcia, D., 2008. Are racial disparities in preterm birth larger in hypersegregated areas? Am. J. Epidemiol. 167, 1295–1304.
Phillips, G.S., Wise, L.A., Rich-Edwards, J.W., Stampfer, M.J., Rosenberg, L., 2009. Income incongruity, relative household income, and preterm birth in the Black Women's Health Study. Soc. Sci. Med. 68, 2122–2128.
Pickett, K.E., Ahern, J.E., Selvin, S., Abrams, B., 2002. Neighborhood socioeconomic status, maternal race and preterm delivery: a case-control study. Ann. Epidemiol. 12, 410–418.
Pickett, K.E., Collins Jr., J.W., Masi, C.M., Wilkinson, R.G., 2005. The effects of racial

density and income incongruity on pregnancy outcomes. Soc. Sci. Med. 60, 2229–2238.
Polednak, A.P., 1991. Black-white differences in infant mortality in 38 standard metropolitan statistical areas. Am. J. Public Health 81, 1480–1482.
Polednak, A.P., 1996. Trends in US urban black infant mortality, by degree of residential segregation. Am. J. Public Health 86, 723–726.
Reardon, S.F., O'Sullivan, D., 2004. Measures of spatial segregation. Sociol. Methodol. 34, 121–162.
Reichman, N.E., Teitler, J.O., Hamilton, E.R., 2009. Effects of neighborhood racial composition on birthweight. Health Place 15, 784–791.
Shaw, R.J., Pickett, K.E., Wilkinson, R.G., 2010. Ethnic density effects on birth outcomes and maternal smoking during pregnancy in the US linked birth and infant death data set. Am. J. Public Health 100, 707–713.
Shihadeh, E.S., Flynn, N., 1996. Segregation and crime: the effect of black social isolation on the rates of black urban violence. Soc. Forces 74, 1325–1352.
Subramanian, S.V., Acevedo-Garcia, D., Osypuk, T.L., 2005. Racial residential segregation and geographic heterogeneity in black/white disparity in poor self-rated health in the US: a multilevel statistical analysis. Soc. Sci. Med. 60, 1667–1679.
Suglia, S.F., Staudenmayer, J., Cohen, S., Enlow, M.B., Rich-Edwards, J.W., Wright, R.J., 2010. Cumulative stress and cortisol disruption among black and hispanic pregnant women in an urban cohort. Psychol. Trauma 2, 326–334.
Vinikoor, L.C., Kaufman, J.S., MacLehose, R.F., Laraia, B.A., 2008. Effects of racial density and income incongruity on pregnancy outcomes in less segregated communities. Soc. Sci. Med. 66, 255–259.
Walton, E., 2009. Residential segregation and birth weight among racial and ethnic minorities in the United States. J. Health & Soc. Behav. 50, 427–442.
White, K., Borrell, L.N., 2011. Racial/ethnic residential segregation: framing the context of health risk and health disparities. Health Place 17, 438–448.
White, M.J., 1983. The measurement of spatial segregation. Am. J. Sociol. 88, 1008–1018.
Williams, D.R., Collins, C., 2001. Racial residential segregation: a fundamental cause of racial disparities in health. Public Health Rep. 116, 404–416.
Williams, D.R., Mohammed, S.A., 2013. Racism and health I: pathways and scientific evidence. Am. Behav. Sci. 57.
Wilson, W., 1987. The Truly Disadvantaged: the Inner City, the Underclass, and Public Policy. University of Chicago Press, Chicago.
Yang, T.C., Matthews, S.A., 2015. Death by segregation: does the dimension of racial segregation matter? PLoS One 10, e0138489.
Yankauer, A., 1950. The relationship of fetal and infant mortality to residential segregation. Am. Soc. Rev. 15.
Yankauer, A., Allaway, N.C., 1958. The relation of indices of fetal and infant loss to residential segregation: a follow-up report. Am. Soc. Rev. 573–578.